

PECST419

CYBER ETHICS, PRIVACY AND LEGAL ISSUES

Module 1

Syllabus:

Fundamentals of Cyber Law and Cyber Space:- Introduction to cyber law, Contract aspects in cyber law, Security aspects of cyber law, Intellectual property aspects in cyber law and Evidence aspects in cyber law, Criminal aspects in cyber law, Need for Indian cyber law

Cyberspace- Web space, Web hosting and web development agreement, Legal and Technological Significance of domain Names, Internet as a tool for global access.

1. Introduction to cyber law

Cyber Law

- Cyber law (also known as Internet law or IT law) is a set of legal rules that manage how people and organizations use computers, the internet, and electronic devices.
- It covers anything you can do online or with digital technology.

Major Areas Covered Under Cyber Law

- **Cybercrimes**
Laws relating to crimes committed using computers or the internet, such as hacking, phishing, cyberstalking, online harassment, and financial fraud.
- **Data Protection and Privacy**
Rules that safeguard personal data of individuals and ensure it is collected and used responsibly.
- **Intellectual Property Rights (IPR)**
Protection for digital content, software, online publications, and creative works such as music, videos, and graphics.
- **Electronic Commerce (E-commerce)**
Regulations that govern online transactions, digital contracts, and consumer protection in online marketplaces.
- **Digital Signatures and Authentication**
Legal validity of electronic signatures and methods used to verify identity in online communications.
- **Cybersecurity Regulations**
Policies and standards that organizations must follow to protect their systems and networks from cyber threats.

Key Objectives & Importance (Purpose)

The main objective of cyber law is to create a **safe and regulated digital environment**. It aims to protect people's rights, ensure security, and establish legal consequences for misuse of technology.

1. To prevent and punish cybercrimes

This objective ensures that illegal online activities such as hacking, fraud, identity theft, and cyberbullying are identified, controlled, and punished under the law. It discourages criminals and protects users from harm.

2. To protect individual privacy and data

Cyber law safeguards personal information shared online, such as passwords, bank details, and private messages. It ensures that organizations handle data responsibly and prevents misuse, leakage, or unauthorized access.

3. To regulate online activities in a fair and secure manner

This ensures that all digital interactions—whether on social media, websites, or online platforms—follow legal rules. It prevents misuse of technology, promotes ethical behavior, and provides guidelines for safe online communication.

4. To promote trust in digital transactions

Cyber law gives legal validity to electronic documents, digital signatures, and online payments. It builds confidence among users and businesses, ensuring that online transactions are safe, reliable, and legally recognized.

5. To ensure a safe digital environment for users, businesses, and governments

This objective focuses on creating a secure cyberspace by enforcing cybersecurity standards, preventing cyberattacks, and protecting digital infrastructure. It helps maintain smooth functioning of online systems for everyone.

2. Contract aspects in cyber law

- Cyber law deals with how contracts are created, validated, and enforced in the online environment.
- Since many transactions today happen digitally like online shopping, banking, subscriptions, and service agreements.
- cyber law provides legal rules to ensure these contracts are valid and secure.
- Below are the major contract-related aspects under cyber law:

1. Legal Recognition of Electronic Contracts (E-Contracts)

Cyber law gives legal validity to contracts formed through electronic means such as emails, websites, or digital forms.

- E-contracts are treated the same as traditional paper contracts.
- Agreements made through “click to accept,” email exchanges, or online forms are enforceable by law.

2. Offer and Acceptance in Online Contracts

Even in the digital world, a contract must have a valid offer and acceptance.

- Clicking “I Agree,” “Submit,” or “Buy Now” counts as valid acceptance.
- Automated responses or digital confirmations serve as proof of agreement.

3. Digital Signatures and Electronic Signatures

Cyber law recognizes digital signatures and e-signatures as legally valid.

- They authenticate the identity of the parties.
- They ensure the integrity of the contract.
- Digital signatures prevent tampering and add security.

4. Authentication and Verification of Parties

Cyber law ensures that the parties entering a contract are genuine.

- Use of OTPs, passwords, digital certificates, or Aadhaar-based e-signing helps verify identities.
- This reduces fraud and impersonation.

5. Electronic Records as Evidence

Cyber law allows electronic documents to be used as legal evidence.

- Emails, digital receipts, logs, and chat records can support contract disputes.
- This is important for proving offer, acceptance, and terms.

6. Formation of Different Types of Online Contracts

Cyber law covers various forms of e-contracts, such as:

- **Click-wrap agreements** (e.g., “Agree and Continue”)
- **Browse-wrap agreements** (terms available on website)
- **Shrink-wrap agreements** (software agreement upon installation)
- Service and subscription agreements

7. Consumer Protection in E-Contracts

Cyber law protects consumers in online transactions by:

- Preventing unfair terms and hidden conditions
- Protecting buyer rights in case of fraud
- Ensuring transparency in pricing, return policies, and warranties

8. Jurisdiction Issues in Online Contracts

Since parties may be in different states or countries, cyber law addresses **which court** has the authority to handle disputes.

- Terms and Conditions often include a “jurisdiction clause” to clarify this.

9. Security and Confidentiality of Online Transactions

Cyber law requires organizations to safeguard data exchanged during online contracts.

- Encryption, secure payment gateways, and privacy policies are mandatory.
- This helps protect sensitive information from theft or misuse.

10. Liability and Enforcement

Cyber law defines the responsibilities of each party and the consequences of violating online contract terms.

- Service providers, sellers, and buyers are bound by the same legal obligations as in physical contracts.

Types of Electronic Agreements

Common forms of online contracts include:

- **Click-wrap Agreements:** The user must explicitly click an "I Agree" or "Accept" button to consent to the terms and conditions. Courts generally uphold these due to the explicit action of consent.
- **Browse-wrap Agreements:** Terms are usually provided via a hyperlink (e.g., in the website footer), and use of the website is deemed as acceptance. These are more legally contentious as explicit consent is not obtained.
- **Shrink-wrap Agreements:** The terms are included inside the product packaging (common with software). The user accepts the terms by opening the package or installing the software

Challenges and Considerations

Despite the robust framework, challenges exist:

- **Jurisdiction:** Determining which country's or state's laws apply when parties are in different locations can be complex.
- **Security:** E-contracts are vulnerable to cyber threats like hacking or data breaches, necessitating secure platforms and encryption.
- **Authentication:** Verifying the true identity and capacity of the contracting parties online can be difficult.

3. Security aspects of cyber law

- The main goal of cyber law security is to keep digital information Confidential, Intact, and Available.
- It achieves this through rules that define crimes, enforce safe practices, and manage security incidents.

1. Defining and Punishing Cybercrimes

Cyber law makes specific harmful digital activities illegal and punishable. This discourages people from committing these acts and provides justice for victims.

Common Cybercrimes Penalized by Law:

- **Hacking:** Getting into a computer system or network without permission.
- **Malware Attacks:** Creating and spreading viruses, ransomware, or worms that damage or disrupt systems.
- **DDoS Attacks:** Overwhelming a website or online service with traffic to crash it (Denial of Service).
- **Phishing & Identity Theft:** Stealing personal information (like passwords or bank details) through deceptive emails or websites.
- **Data Theft:** Illegally copying or moving sensitive information.
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2. Setting Mandatory Security Rules for Businesses

Cyber law forces organizations that handle your data to protect it properly. They can't just ignore security risks.

- **"Reasonable Security Practices":** Businesses must put in place standard, sensible security measures. This includes things like:
 - Using firewalls (digital walls) to block attacks.
 - Using encryption (scrambling data) to protect sensitive information like bank details.
 - Requiring strong passwords or multi-factor authentication (e.g., a password *plus* a code sent to your phone).
 - Conducting regular security checks or "audits".
- **Industry Standards:** Often, laws point to recognized international security standards like ISO/IEC 27001 that companies must follow.

3. Handling Security Breaches

What happens when a security breach occurs? Cyber law makes sure companies don't hide it.

- **Mandatory Reporting:** Many laws require companies to report data breaches quickly to government authorities. In the EU under GDPR, this is often within 72 hours of discovery. In India, reporting must happen within six hours to CERT-In.
- **Notifying Victims:** Companies usually have a legal obligation to inform the people whose data was compromised.
- **Forensics and Logs:** Cyber law often requires businesses to keep system logs (records of activity) for a certain time, which helps investigators figure out what happened during an attack.

4. Protecting Important National Systems

Cyber law provides special protections for "Critical Information Infrastructure"—systems essential for a country's function.

- **Examples:** Power grids, banking systems, military networks, and transportation systems.
- **Goal:** To protect these from large-scale attacks that could harm national security or the economy.

5. Accountability and Penalties

If companies or individuals break these security laws, there are consequences. Cyber law enforces fines, legal action, and even jail time for negligence or malicious intent.

4. Intellectual property aspects in cyber law and Evidence aspects in cyber law

In the digital world, creative works (software, music, videos, graphics, designs, photos, websites, books, etc.) can be copied or shared easily. Cyber law provides rules that protect these creations through Intellectual Property Rights (IPR).

Key IPR Aspects in Cyber Law

1. Protection of Copyright in Digital Content

Cyber law ensures that digital creations like software, e-books, articles, images, and videos cannot be copied or distributed without permission.

- Covers issues like piracy, illegal downloading, and unauthorized streaming.
- Helps creators maintain ownership and control.

2. Software Piracy Prevention

Copying, distributing, or using software without a license is illegal. Cyber law protects:

- Application software
- Operating systems
- Source code
- Mobile apps

Businesses using pirated software can face penalties.

3. Trademark Protection on the Internet

Brands, logos, names, and domain names are protected under cyber law. It prevents:

- **Cybersquatting:** Registering someone else's brand as a domain name.
- **Passing off:** Using similar names or logos to confuse customers.

- Fake online stores misusing trademarks.

4. Patents for Digital Inventions

- Innovations like new algorithms, software-based processes, and technological systems may be patentable.
- Cyber law ensures protection against duplication or misuse of patented digital technologies.

5. Protection Against Online Plagiarism

- Copying content from websites, blogs, e-books, or social media without permission is a violation.
- Cyber law helps creators file claims against plagiarism and copyright theft.

6. Licensing and Terms of Use

E-contracts often define how digital content can be used. Cyber law enforces these licenses, such as:

- Creative Commons licenses
- Software licenses
- Website usage agreements

7. Digital Rights Management (DRM)

Cyber law supports technological tools that prevent unauthorized copying or sharing of digital media.

Evidence Aspects in Cyber Law (Cyber Evidence / Digital Evidence)

Cyber law also deals with how electronic information is used as evidence in legal proceedings. Since crimes today often involve computers or mobile phones, digital evidence is extremely important.

Key Evidence Aspects in Cyber Law:

1. Legal Recognition of Electronic Records

Cyber law gives electronic records the same legal status as paper documents. Electronic records include:

- Emails
- Text messages
- Digital logs
- Computer records
- CCTV footage
- Call records
- Metadata
- Social media posts

2. Admissibility of Digital Evidence

For electronic evidence to be accepted in court, it must be:

- Authentic
- Reliable
- Untampered
- Properly collected and preserved

Many laws require a certification process (like Section 65B certificate in India).

3. Chain of Custody

This refers to documenting who handled the digital evidence, when, and where. It ensures evidence is not altered or corrupted. Proper chain of custody is mandatory for admissibility.

4. Forensics and Log Preservation

Cyber law often requires organizations to keep:

- Server logs
- Login records
- Transaction trails
- System activity reports

These logs help in investigation and prosecution of cybercrimes.

5. Authenticity and Integrity of Evidence

Cyber law uses methods like:

- Hash values
 - Digital signatures
 - Timestamps
- to prove the evidence was not changed.

6. Search and Seizure of Digital Devices

Law enforcement agencies have powers (with warrants) to seize:

- Computers
- Mobile phones
- Hard drives
- Cloud data for investigation.

7. Expert Testimony

Cyber forensic experts are often required to explain technical evidence such as:

- IP addresses
- Malware behavior
- Server logs
- Data recovery

Their reports help the court understand complex digital evidence.

8. Preservation Orders

Courts or investigating agencies can order companies to preserve specific digital records for investigation or litigation.

5. Criminal aspects in cyber law

Cyber law defines various digital activities as criminal offenses and provides punishments for people who misuse technology. Because crimes can be committed anonymously and across borders, cyber law ensures there are clear rules to identify, prevent, investigate, and punish such offenses.

Categories of Cybercrimes

Cyber law typically addresses two main categories of crimes:

- **Computer as a Target:** Crimes where the digital system itself is the primary target.
- **Computer as a Weapon:** Traditional crimes (like fraud or theft) that are committed or facilitated using a computer or the internet.

Key Criminal Offenses

Cyber law statutes, such as the Indian Information Technology Act, 2000 or the US Computer Fraud and Abuse Act (CFAA), criminalize specific acts:

- **Hacking/Unauthorized Access:** Illegally accessing or attempting to access a computer system, network, or data without permission.
- **Malware Distribution:** Creating, using, or distributing viruses, worms, ransomware, or Trojan horses designed to damage, disrupt, or gain unauthorized access to computer systems.
- **Denial-of-Service (DoS) Attacks:** Overwhelming a network or server with excessive traffic to make an online service unavailable to legitimate users.
- **Identity Theft and Impersonation:** Fraudulently using another person's personal information, passwords, or electronic signatures to commit fraud or for personal gain.
- **Online Fraud and Financial Crimes:** Deceiving individuals online for financial gain through activities like phishing, online scams, credit/debit card fraud, or business email compromise.
- **Cyberstalking and Harassment:** Using electronic communications (email, social media, etc.) to persistently follow, threaten, or intimidate a person.
- **Publishing Obscene or Harmful Content:** Transmitting or publishing sexually explicit material, child pornography (which carries severe penalties), or defamatory content online.

- **Tampering with Computer Source Documents:** Intentionally concealing, destroying, or altering source code that is legally required to be maintained.
- **Cyberterrorism:** Using digital means to cause mass disruption, threaten national security, or strike terror in people (punishable by severe penalties, including life imprisonment in some jurisdictions).

6. Need for Indian cyber law

With the rapid growth of the internet, digital commerce, online communication, and electronic services, India required a strong legal framework to regulate online behavior and protect users. Indian cyber law was introduced to ensure safety, security, and trust in the digital environment.

Need for Indian Cyber Law :

1. Growth of Internet and Digital Communication

As millions of Indians began using computers, smartphones, email, social media, and online services, it became essential to create laws that govern digital activities. Cyber law ensures that online interactions are safe and legally regulated.

2. Increase in Cybercrimes

Cybercrimes such as hacking, fraud, phishing, data theft, cyberbullying, and online scams have grown rapidly. India needed cyber laws to:

- Define cybercrimes
- Provide punishments
- Help police investigate and prevent such offenses

3. Legal Recognition for Electronic Transactions

To support e-commerce, online banking, digital signatures, and electronic records, India needed a law giving them legal validity. The Information Technology Act, 2000 made e-contracts, e-signatures, and electronic records legally recognized.

4. Protection of Personal Data and Privacy

With companies collecting huge amounts of personal information, India needed laws to ensure:

- Data is used responsibly
- Users' privacy is protected
- Organizations follow security standards

Cyber law helps prevent misuse of personal and sensitive data.

5. Promoting E-Commerce and Digital Payments

Online shopping, digital wallets, and online banking needed a secure legal foundation. Cyber law promotes trust by ensuring that digital transactions are safe and enforceable.

6. Securing Critical Information Infrastructure

Important systems like:

- Banking networks
- Telecom
- Power grids
- Government databases

Must be protected from cyberattacks. Cyber law defines special protection measures for these critical systems.

7. Regulation of Online Content and Behavior

India needed cyber laws to control:

- Cyberbullying
- Online hate speech
- Fake news
- Obscene or harmful content
- Misuse of social media

It helps maintain social order and protects citizens online.

8. International Collaboration

Cybercrimes often involve multiple countries. India needed cyber laws to cooperate with global agencies and track cross-border cybercriminals.

9. Supporting Digital India Initiatives

As India moves toward a digital economy, with projects like:

- Digital payments
- E-governance
- Online services

strong cyber laws are needed to ensure these systems run smoothly and securely.

10. Strengthening National Security

Cyber threats like espionage, terrorism, and cyber warfare require laws that empower authorities to protect national security. Cyber law ensures strong defense against digital attacks.

7. Cyberspace- Web space

Cyberspace

- Cyberspace refers to the virtual environment created by computers, digital networks, and the internet.
- It is the “digital world” where people communicate, share information, and perform online activities.

features of cyberspace:

- Exists electronically, not physically
- Includes computers, networks, servers, and online platforms
- Supports activities like communication, social media, gaming, banking, and e-commerce
- Accessible from anywhere globally

Web Space

Web space is a part of cyberspace. It refers specifically to the portion of the internet that contains websites and web pages.

It includes:

- Websites
- Web pages
- Blogs
- Online stores
- Search engines
- Online content hosted on the World Wide Web

Difference Between Cyberspace and Web Space

Cyberspace	Web Space
The entire digital world created by computers and networks	A part of cyberspace that contains websites
Includes emails, apps, social media, games, databases, networks	Refers only to content on the World Wide Web
Broader concept	Narrower concept

8. Web hosting and web development agreement

Web Development Agreement

- A Web Development Agreement is a legal contract between a web developer or development company and a client who wants a website or web application created.
- It defines the scope, timeline, cost, and responsibilities for designing, developing, and delivering a website or web application.

Components of a Web Development Agreement:

1. **Scope of Work (SOW):**
 - Design, development, coding, database integration, testing, and deployment
2. **Timeline and Delivery:**
 - Project start date, milestones, and final delivery date
3. **Payment Terms:**
 - Total cost, installment structure, penalties for delays
4. **Ownership and Intellectual Property:**
 - Client usually owns the website content and code after full payment
 - Developer may retain the right to reuse non-exclusive components
5. **Confidentiality:**
 - Developer must keep client information, data, and source code confidential
6. **Maintenance and Support:**
 - Post-deployment support and bug fixes
7. **Termination Clause:**
 - Conditions for terminating the agreement before project completion
8. **Liability and Warranties:**
 - Developer may provide warranty for a limited period
 - Limitations on damages for defects or delays

Web Hosting Agreement

- A Web Hosting Agreement is a legal contract between a web hosting service provider and a client who wants to host their website on the provider's servers.
- It defines the terms and conditions for hosting, maintaining, and managing a website on the internet.

Components of a Web Hosting Agreement:

1. **Services Provided:**
 - Server space, bandwidth, storage, database access, email services, etc.
2. **Service Level Agreements (SLA):**
 - Uptime guarantees (e.g., 99.9%)
 - Response time for technical support
3. **Payment Terms:**
 - Hosting fees, renewal charges, additional costs
4. **Security and Data Protection:**
 - Backup policies
 - Responsibility for data loss or breaches
5. **Ownership and Intellectual Property:**
 - Client retains ownership of website content
 - Provider may only manage storage and access
6. **Termination Clause:**
 - Conditions under which either party can terminate the agreement
7. **Liability and Indemnity:**
 - Limits on the provider's responsibility for downtime, data loss, or hacking

Web Hosting Agreement VS Web Development Agreement

Web Hosting Agreement	Web Development Agreement
Focuses on hosting a website on a server	Focuses on creating/designing a website
Provider manages server space, uptime, and security	Developer delivers the website code, design, and functionality
Ongoing service agreement	Project-based agreement
Client owns content; provider manages storage	Ownership of website usually transfers after payment

9. Legal and Technological Significance of domain Names

- A domain name is the unique address used to identify a website on the internet (e.g., www.example.com).
- It is crucial both legally and technologically because it represents identity, access, and ownership online.

Legal Significance of Domain Names

a) Intellectual Property Protection

- Domain names are often linked to brand names or trademarks.
- Legal disputes can arise over cybersquatting (registering someone else's brand name as a domain) or typosquatting (using similar names to mislead users).
- Laws and regulations protect trademark owners from unauthorized use of their domain names.

b) Business Identity and Reputation

- A domain name serves as a company's digital identity.
- Legal protection ensures businesses maintain their online reputation and avoid impersonation or fraud.

c) Dispute Resolution

- Courts and domain registration authorities (like ICANN or NIXI in India) resolve conflicts over domain ownership.
- Legal frameworks prevent misuse and ensure fair allocation of domain names.

d) Enforceability of Contracts

- Domain names can be part of licensing agreements, website agreements, or intellectual property contracts.
- Ownership and usage rights are legally enforceable.

Technological Significance of Domain Names

a) Easy Access to Websites

- Domain names provide a user-friendly way to access websites instead of remembering IP addresses.
- Example: Instead of typing 192.168.1.1, users type www.example.com

b) Network Addressing

- Domain names are mapped to IP addresses using the **Domain Name System (DNS)**.
- DNS translates human-readable names into numeric addresses understood by computers.

c) Internet Navigation and Searchability

- Domain names help in organizing and locating websites efficiently.
- They are essential for search engines, URL structuring, and web hosting.

d) Security Implementation

- Domain names are associated with SSL certificates and HTTPS, which ensure secure online communication.
- They help prevent phishing and malicious attacks by verifying authentic websites.

e) Branding and Digital Marketing

- Memorable and relevant domain names improve online visibility and marketing.
- They help users identify, trust, and remember a business or service online.

10. Internet as a tool for global access

The internet serves as a powerful tool for global access by creating a vast, interconnected network that transcends geographical boundaries, enabling seamless and instant exchange of information, communication, and services across the world.

The internet facilitates global access through several key mechanisms:

- **Interconnected Infrastructure:** It functions as a "network of networks" made up of physical cables (fiber optic, copper wires) and wireless connections (Wi-Fi, 4G/5G) that link billions of devices worldwide.
- **Standardized Protocols:** Devices communicate using universal rules and protocols (like TCP/IP and HTTP) that ensure data can be transmitted and interpreted correctly by any connected machine, regardless of its location or hardware.
- **World Wide Web (WWW) and Services:** The internet provides the infrastructure for services like the WWW, email, social media, and streaming, allowing users to access a vast array of content and interact through user-friendly applications.

Internet as a Tool for Global Access

The internet is a global network that connects millions of computers and devices worldwide. It acts as a powerful tool for accessing information, services, and communication across borders.

1. Access to Information

- The internet provides instant access to vast amounts of information from anywhere in the world.
- Examples: Online encyclopedias, research articles, news websites, government portals.
- Users can learn, research, and stay updated globally.

2. Communication Across Borders

- Enables instant communication through email, social media, messaging apps, and video calls.
- Businesses, students, and individuals can interact with people across countries without physical travel.

3. E-Commerce and Business Expansion

- Companies can sell products and services internationally through online platforms.
- Small businesses can reach a global customer base, making the internet a tool for global trade and economy.

4. Online Education and Collaboration

- Students and professionals can access online courses, webinars, and collaborative platforms.
- Universities and institutions offer distance learning programs, enabling global education access.

5. Access to Government and Health Services

- Citizens can access government services online, such as taxes, licenses, or certificates.
- Telemedicine and online health platforms provide medical advice and consultations globally.

6. Cultural Exchange and Awareness

- The internet allows sharing of cultures, languages, and ideas across countries.
- People can connect with different communities, promoting global understanding.

7. Social and Political Awareness

- Citizens can participate in global discussions, campaigns, and social movements.
- News, blogs, and forums provide worldwide perspectives on political and social issues.